

MUSICAL BRAINS

Playing music involves multiple parts of the brain. Studies comparing the brains of professional musicians and amateurs revealed that professional musicians had a greater volume of gray matter in brain areas related to motor, auditory, and visual-spatial reasoning. The study's findings show how the brain undergoes structural adaptations in response to the environment (dedicating hours to repetitive rehearsals with an instrument).



THE HIPPOCAMPUS IN AN ADULT BRAIN MAKES AN ESTIMATED 700 NEW NEURONS EVERY DAY



Genes versus environment

People are born with a DNA “template” inherited from their parents (see pp.56–57): this is the “nature” element influencing the brain’s activities, such as cognitive ability and behavior. Throughout a person’s life, though, their networks of neurons (see pp.26–27) can adapt and change in response to physical and social experiences (“nurture”). Environmental influences, if strong and sustained, can alter brain structures and also influence the way that genes work—a process known as epigenetic change (see opposite).

Nature and Nurture

The two fundamental influences on the brain, “nature” and “nurture,” are sometimes seen as opposing forces. However, there is a dynamic interplay between them that goes on throughout a person’s life.

NATURE



CHROMOSOMES

We inherit our chromosomes, which contain our DNA, from our parents (see pp.56–57). It’s the chromosomes that, at the point of fertilization, determine the chromosomal sex of an embryo (XX for female and XY for male). Chromosomal abnormalities can also cause disease or developmental problems.



DNA

Some psychological traits, such as the tendency to develop depression, have been linked to particular genes—but they usually involve dozens or even hundreds of the genes acting together. The more of those genes a person inherits, the more likely they are to develop that trait.

WHEN DO EPIGENETIC CHANGES HAPPEN?

Epigenetic changes can be induced by environmental factors at any point in a person’s life, from development in utero to old age.